

Service Contracts

A service contract defines the operations that other services may call. It specifies the input, output, and possible errors while hiding the internal implementation details.

3.1 User Service

Operation	Input	Output	Errors
registerUser	Name, email, password	User registration result	Invalid input; email already registered
loginUser	Email, password	Authentication result	Invalid credentials; account unavailable
getUser	User reference	User information	User not found

3.2 Catalogue Service

Operation	Input	Output	Errors
searchFoodItems	Search query; optional category	Matching food items	Invalid search query
getFoodItem	Food item reference	Food item details and price	Food item not found
checkItem	Food item reference; quantity	Item availability result	Food item not found; item unavailable
getVendor	Vendor reference	Vendor information	Vendor not found

3.3 Order Service

Operation	Input	Output	Errors
addToCart	Food item reference; quantity	Updated cart	Item unavailable; invalid quantity
placeOrder	Customer reference; cart; fulfilment details	Order confirmation and status	Empty cart; item unavailable; payment failure
cancelOrder	Order reference	Updated order status	Order not found; cancellation not allowed
trackOrder	Order reference	Current order status	Order not found

3.4 Payment Service

Operation	Input	Output	Errors
makePayment	Order reference; amount; payment details	Payment result	Payment declined; invalid payment details; payment service unavailable

Contract boundary: These contracts expose only the operations, inputs, outputs, and error cases available to callers. Internal implementation details are hidden.